

Task 1: “How Do Big Systems Stores Data?”

Large-scale applications like Instagram, Uber, and Amazon utilize SQL and NoSQL databases for their specific performance, scalability, and data consistency requirements.

- Instagram:

Instagram uses PostgreSQL for relational data, sharding and replication for scalability, and Apache Hive for long-term storage and analytics for efficient querying and analysis of large datasets.

- Uber:

Uber uses schemaless, a MySQL-based distributed database system, Docstore for high-throughput, low-latency operations, and incorporates Riak and Cassandra for high availability and fault tolerance.

- Amazon:

Amazon uses Amazon Aurora, a cloud-native relational database, for its primary database and data replication, and has migrated from Oracle to AWS solutions for cost reduction and performance improvement.

Summary: SQL vs. NoSQL in Large Applications

Large-scale applications typically use **both SQL and NoSQL databases**, each serving different roles depending on the system’s needs.

- **Instagram** relies on **SQL (PostgreSQL)** for storing structured data like user profiles and relationships, but uses **big data tools** like Apache Hive for analytics over massive image and usage data.
- **Uber** builds its systems primarily on **MySQL**, but in a distributed, custom-engineered form called *Schemaless* and *Docstore*. For other high-performance needs like location tracking and messaging, they incorporate **NoSQL databases** such as **Riak** and **Cassandra**.
- **Amazon** uses a relational database called **Amazon Aurora**, which supports SQL-based queries and is highly scalable in the cloud. For other use cases—like product catalogs, shopping carts, or user sessions—it uses **NoSQL** databases like **DynamoDB**, which offer rapid scaling and flexible data models.

Use of SQL Server in Large Applications:

Microsoft SQL Server, a robust relational database system, is not commonly used in large-scale internet-based applications like Instagram, Uber, or Amazon, but is prevalent in Microsoft-integrated enterprise environments.

Task 2: “Do Video Game USe Database?”

How Online Games Store Player Data?

Online games store vast amounts of player data, including profiles, progress tracking, match histories, inventory systems, and leaderboards. Efficiently managing this data involves using Relational Databases (SQL) for structured data, NoSQL Databases for unstructured data, and in-memory caching for rapid access to frequently used data, reducing latency and ensuring consistency in player accounts and transaction records.

Example of a Game Using a Relational Database:

MMORPGs often use relational databases to manage complex relationships between players, items, and mechanics. For example, Dragonfly uses PostgreSQL to store unique identifiers, usernames, progression stats, item IDs, descriptions, and ownership details for players, items, and quests. These structured relationships ensure data integrity and support complex queries.

Summary:

Online games use a hybrid approach of SQL Databases for structured, relational data, NoSQL Databases for unstructured data, and Caching Systems for memory storage, ensuring seamless, real-time experiences for millions of players worldwide.